

Investigation of sustainable operation oriented- economic, process and environment based multi-criteria optimization of large scale methanol production plant

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ABSTRACT

Methanol is one of the main raw materials and a building block for many essential chemicals in the industry, and it is widely produced from the synthesis gas. It is used as a fuel, solvent, inhibitor, antifreeze agent, and in producing a variety of chemicals including olefins and paraffin. The global demand for methanol is 70 million tons per year and it is estimated to grow by a factor of 6–8% per year so it is necessary to develop a methanol production process, which is economical, and environment friendly. The multi-objective optimization (MOO) approach is used to optimize the large scale Lurgi two-step methanol synthesis process using the non-dominated sorting genetic algorithm-II (NSGA-II) to find the Pareto front solutions involving the process, economic, and environmental-based objectives. Methanol production rate, total energy consumption, total annual CO₂ emissions, and profit are considered as objectives. The five decision variables include syngas molar flow, reactor feed pressure, reactor 1 temperature, reactor 2 temperature, and by-product mass flow in the methanol distillation column. Two constraints are set on ethanol mole fraction in the methanol stream, and methanol mole flow in water to the treatment stream. Three optimization cases (two bi-objective and one tri-objective) are developed. In the Case 1 study, CO₂ emissions decreased by 65.83%, the methanol production rate increased by 10.11%, and total energy was reduced by 57.55%. The Pareto ranking is carried out using the Preference Ranking On the Basis of Ideal-average Distance (PROBID) method and the best optimal solution is reported for all cases.

1. Introduction

1.1. Motivation

Methanol or methyl alcohol with the chemical formula of CH₃OH is a light, flammable, volatile, and colorless liquid (Cheng, 1994). Methanol is an essential raw material in producing a variety of chemicals such as olefins and paraffin. Also, it is used as fuel, solvent, inhibitor, antifreeze, or substrate (Anicic et al., 2014; Rafiee, 2020). The global demand for methanol is 70 million tons per year and it is estimated to grow by a factor of 6–8% per year so it is necessary to develop a methanol production process that is economical and environment-friendly (Rafiee, 2020). Methanol is produced from various sources, such as natural gas, biomass, and coal. Usually, methanol production plants that use natural

gas as a feedstock contain three main sections. The first section is the synthesis gas generation section where natural gas hydrocarbons are converted into syngas (a mixture of hydrogen, carbon monoxide, carbon dioxide, methane, and impurities such as nitrogen and sulfur compounds) (Deng et al., 2015). Secondly, the methanol synthesis section where methanol production reactions occur within a recycle loop to generate a high amount of methanol. Finally, methanol purification takes place in distillation columns to remove the dissolved gases and the by-products and to achieve the desired methanol purity (Bertau et al., 2014). The main steps in the methanol production process are the methanol synthesis reaction (Eq. (1)) and the reverse water gas shift reaction (Eq. (2)), carried out on a surface of copper zinc oxide catalyst with alumina oxide at high pressure of 50–100 bar and temperature of 200–300 °C (Sheldon, 2017; Bukhtiyarova et al., 2017). Two additional

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side reactions resulted in the by-product formation of ethanol (Eq. (3)) and dimethyl ether (DME) (Eq. (4)) (AspenTech, 2018).



1.2. Literature review

Different methanol generation processes are used in the industry. The conventional commercial processes usually operate with a capacity of less than 3000 tons per day and the large scale process designs produce more than 5000 tons per day of methanol (Bertau et al., 2014). The Imperial Chemical Industries (ICI) Johnson Matthey low pressure methanol (LPM) process is one of the leading industrial methanol processes worldwide, producing more than 30 million metric tons of methanol per year (Bertau et al., 2014; Sheldon, 2017; Bozzano and Manenti, 2016). The conventional Lurgi methanol production process is an isothermal process developed by Lurgi where methanol is produced by a catalytic reaction in two water cooled reactors. The conventional Lurgi process capacity is lower than 3000 tons per day of methanol. For a larger capacity of more than 5000 tons per day, the Lurgi MegaMethanol Process and Lurgi GigaMethanol Process are used (Bertau et al., 2014; AirLiquide, 2022). The Lurgi MegaMethanol Process is specially designed for producing methanol in large quantities with low production cost and it is usually used for a capacity greater than 5000 tons per day (Adil and Rao, 2022). The Lurgi GigaMethanol Process is designed to produce up to 10,000 tons per day in a single train (AirLiquide, 2022). The Haldor Topsøe Process is one of the processes, that used natural gas as a feedstock to produce methanol, consisting of two steps of reforming process to generate synthesis gas (Marie-Rose et al., 2011). This process is appropriate for small scale and large scale capacities up to 10,000 tons per day (Bertau et al., 2014; Bukhtiyarova et al., 2017). Mitsubishi Heavy Industries (MHI) and Mitsubishi Gas Chemical company (MGC) collaborated to develop new technology in methanol production to save energy, reduce investment and minimize operating cost with higher efficiency. The developed technology used a new isothermal converter known as the MGC/MHI super converter. (Bertau et al., 2014; Bozzano and Manenti, 2016).

Chemical industrial production processes are complex in nature and involve various factors and criteria that need to be attended simultaneously to achieve the target goal of production with all the standard operating procedures to be followed. To achieve this goal, a MOO technique that considers different conflicting objectives along with the decision variables and constraints is beneficial (Gujarathi et al., 2023a). MOO techniques have shown successful results in terms of analysis of various objectives and outcomes in the variety of disciplines and sectors such as biochemical (Patel et al., 2023a, 2023b; Gujarathi et al., 2023b), chemical (Joo et al., 2024), oil and gas (Thafseer et al., 2021; Panda and Ramteke, 2020; Al-Aghbari et al., 2022), polymer (Gujarathi and Babu, 2013; Ramteke and Gupta, 2008), metallurgical (Mahanta and Chakraborti, 2018), biodiesel (Deshpande et al., 2022), zeolite production (Tauanov et al., 2018), waste to resource (Lee et al., 2022), explosive waste treatment (Cho et al., 2023).

Chen et al. (2011) optimized and designed a Lurgi methanol synthesis reactor model using the Aspen Plus program. The optimization variables were the inlet temperature of the syngas, cooling water volumetric flow rate, and pressure of the steam drum. The optimization problem was studied to maximize methanol production. The efficiency of methanol production increased by 7.04%. Bayat et al. (2014) developed a MOO problem to maximize selectivity and methanol production in a gas flowing solid fixed bed reactor using the NSGA-II. Stoica et al.

(2015) carried out an optimization study using a genetic algorithm to maximize carbon oxides conversion and production flow rate for a methanol reactor operated using direct and indirect cooling approaches. Two optimization problems were formulated and compared for each cooling technique. The first problem considers no temperature restriction and the second one takes into account the temperature limitation at the inlet of each catalyst bed.

Leonzio (2020) developed an optimization problem to maximize methanol production, yield, carbon dioxide conversion, and selectivity and minimize reaction outlet temperature. The ANOVA analysis and Yates algorithm were used to identify the significant factors that affect the process, which is modelled by CHEMCAD. The authors reported the optimal operating conditions for inlet temperature, reaction pressure, and fraction of carbon to be equal to 493.32 K, 55 bar, and 0.8 respectively. Rafiee (2020) studied an optimization problem to maximize methanol mass production, CO₂ utilization and H₂ conversion in methanol synthesis. It was found that all the studied three objectives were greater when H₂/CO₂ ratio was equal to three compared to when the H₂/CO₂ ratio was equal to two. Borisut and Nuchitprasittichai (2020) studied three different design configurations of methanol production through CO₂ hydrations. The configurations were simulated using Aspen HYSYS and optimized to minimize the production cost. The results showed that the lowest production cost could be obtained by using two reactors in a series configuration and the hydrogen cost has a high impact on the overall methanol production cost.

Adil and Rao (2022) studied the Aspen Plus model for methanol production with Vanden Bussche and Graafs kinetics, and optimized the process for the maximization of production and yield of methanol and maximization of carbon conversion. The authors used a surrogate model which they obtained using a third-order polynomial correlation using a regression fit. Such surrogate models may show limitations in terms of accuracy under a wider range of solutions. However, they also did not consider any multi-objective study. Their process model was also not valid for large-scale production of methanol. Dehghani et al. (2021) developed an optimization study for the first principle-based gas-cooled reactor model to maximize selectivity and methanol production rate using the NSGA-II. Their study was limited only to process-based objectives and did not consider the large-scale production of methanol. The CO₂ emissions analysis was limited to reactor design and configuration and CO₂ emissions analysis was not performed from MOO preceptive. In another study, Seidel et al. (2021) used first principle based model and developed a nonlinear programming optimization for methanol synthesis under periodic operation compared to optimal steady state operation to maximize methanol outlet flow rate and methanol yield. The results showed that increasing hydrogen concentration in the feed has positive impacts on the optimization objectives. However, the author's considered only the process-based objectives. Maksimov et al. (2022) carried out a MOO study to maximize methanol purity and methanol production rate in two different reactor models using NSGA-II. The authors used partial differential equations-based conservation models valid for small-scale methanol production. Their MOO study was also limited to process-based objectives. Considering the observed gaps from cited literature such as the studies using the first principal-based models, studies limited to small-scale production of methanol, studies not considering diverse sets of objective functions, there exist a scope to study large scale methanol production process from the MOO perspective considering process, economics, environment, and energy-based criteria.

1.3. Novelty of work

The novelty of the present study can be summarized through the following points:

- This is the first attempt to carry out MOO study using NSGA-II for large-scale Lurgi two-step methanol production process plant.

- A variety of objectives such as process (methanol production rate, and total energy), economic (profit), and environmental (total annual CO₂ emissions) are considered along with five operation-oriented decision variables and two constraints.
- Two cases of bi-objective optimization and one case of tri-objective optimization are considered using the conflicting set of objectives.
- The PROBIT method of Pareto ranking is utilized to rank the best optimal solution in each case.

The rest of the manuscript is arranged as follows: The second section on methodology consists of process description and problem formulation. In the next section i.e., the third section detailed results analysis of three optimization cases are discussed and in the fourth section, the conclusions of the study are presented followed by the references used.

2. Methodology

2.1. Process description

Large scale Lurgi two-step methanol synthesis process is considered for the MOO analysis in this current study. The Lurgi process is the leading industrial methanol production process worldwide, where Lurgi reactors provide 30% of the methanol global demand. In this production process, higher uniformity of temperature ensures higher yield (Bertau et al., 2014). As shown in Fig. 1, the large scale Lurgi two-step process is fed with sulfur-free syngas at a temperature of 37.8 °C and a pressure of 35.5 bar with a molar composition of 6.8% carbon dioxide, 22.8% carbon monoxide, 67.5% hydrogen, 2.2% methane, and a small amount of inert gases, and water. The syngas feed is pressured to 92.8 bar in the compressor (Comp-1) and a condensate stream is taken off downstream from the compressor intercooler (HEX-3). This gas is then mixed with the recycled syngas from the compressor (Comp-2) reaching a temperature of 58.9 °C. Then heated by the reactor effluent in the heat exchanger (HEX-4) reaching a temperature of 182.2 °C. After that, the syngas enter the tube side of the second converter (Reactor-2). Then syngas passes to the first stage converter i.e., (Reactor-1) which is a boiling water reactor operated at isothermal conditions. The reaction occurs inside tubes packed with catalysts. The second stage reactor (Reactor-2) is a gas-cooled reactor and the reaction occurs in the shell side, which is packed with catalyst. The Reactor-2 is cooled by the

incoming syngas passing through the tube side, which is used to feed the first-stage converter. This cooling process allows higher conversion and removes the excess heat from the process. The product gas leaves the second reactor at a temperature of 227.8 °C and a pressure of 87.9 bar. Then the temperature of the product gas is reduced to 190.5 °C in the exchanger (HEX-7). In the separation part of the process, the product gas enters the drum (Vessel-1). Part of the light gases is recompressed by the compressor (Comp-2) and mixed with the pressured syngas feed. The remaining part of the light gases is scrubbed with water in column (Column-1) to recover the residual methanol and finally, the light ends from column (Column-1) mix with the light ends from column (Column-2) and are used as fuel in the process. The condensed methanol from column (Vessel-1) and the recovered methanol from column (Column-1) flow to the vessel (Vessel-2). Then the product methanol is let down to near atmospheric pressure in a storage tank (Tank-1), where a small vapor purge comes off the tank. The crude methanol from the storage tank (Tank-1), which contains residual methanol, light gases, and heavy products such as ethanol, DME, and other alcohol, is pumped to the light ends column (Column-2), where the methanol is separated from residual light ends. The methanol from the bottom of the light ends column (Column-2) flows to the methanol product column (Column-3), where methanol is taken overhead, and water from the bottom of the column and the side stream purges ethanol and other alcohols. The design and mathematical model details are given in the Appendix Table A1 and Table A2 respectively. The role of each device in Fig. 1 is shown in the Appendix Table A3.

2.2. Problem formulation

A typical optimization problem consists of an objective function that is to be minimized or maximized based on the bounds of decision variables within the limits of constraints if any. In this study, different objective functions based on process, economic, and environmental aspects are identified and studied. The first two objectives selected are based on the process aspects i.e., F1 = maximize methanol production rate (kmol/h), and F2 = minimize total energy (kWh). The Third objective function is F3 = minimize total annual CO₂ emissions (kg/y), and the last objective function is F4 = maximize profit (US\$/y).

For this MOO analysis five decision variables namely syngas molar flow, reactor feed pressure, reactor 1 temperature, reactor 2

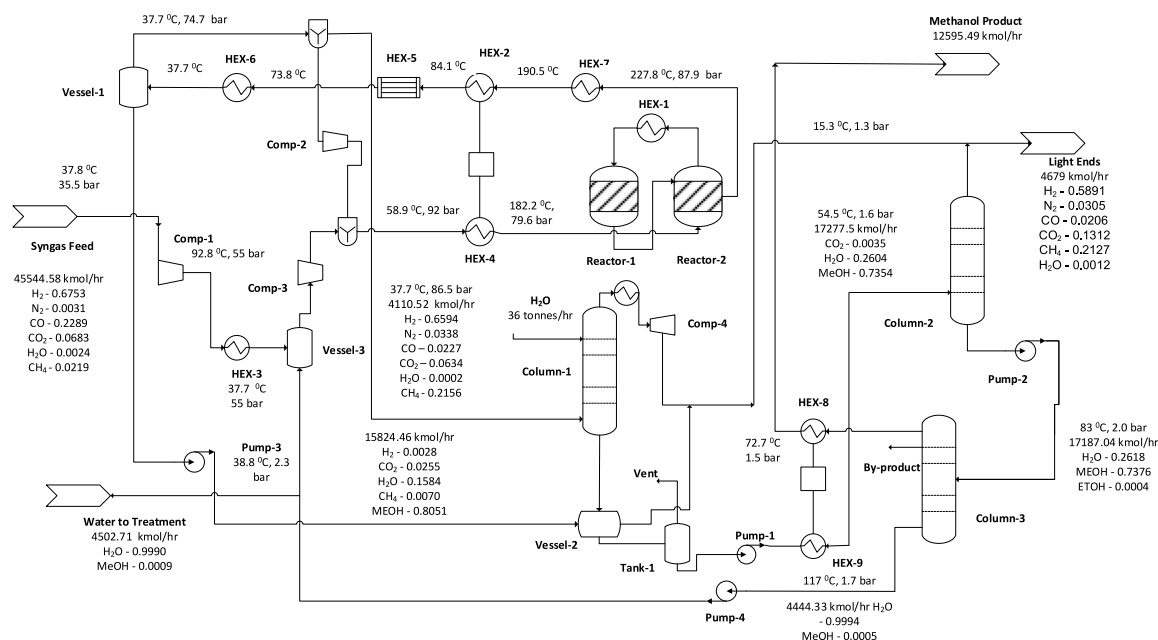


Fig. 1. Process flow diagram of large scale Lurgi two-step methanol synthesis model.

temperature, and the by-product mass flow in the methanol distillation column are considered. The syngas gas molar flow rate has a direct relation with the product formation (through reaction 1 stoichiometry), the methanol product rate (and also the byproduct rates) increases with increasing the syngas flow rate. The conflict between the formation of the main product and the side products can be controlled by controlling the reactor temperature-which are considered as decision variables. The byproduct mass flow is considered as one of the specifications in the separating column, which had shown an impact on the distillate and the bottoms rate of the column, and thus it is considered as a decision variable. The upper and lower bounds of the decision variables are selected based on the detailed sensitivity analysis study and are shown in Table 1 along with the constraints and their limits.

Maximizing the methanol production rate is the first objective that has been considered in this study. One of the objectives considered in this optimization study is the minimization of the total energy consumed in the process, and it can be calculated by the following equation.

$$\text{Total energy} = \sum \text{energy production} - \sum \text{energy consumption} \quad (5)$$

For the above equation (5), the energy and power interchange system consists of two main systems. The first one is an energy-generating system where energy is generated using an energy source and transmitted by an energy path. The second system considers the consumed energy supplied by the energy path. The energy produced consists of energy produced from all -coolers and -condensers. The energy consumed consists of the consumed energies from all -pumps, -compressors, -heaters and -reboilers. For the details of equipment and its role, please refer to Fig. 1 and Table A3 in the appendix. The analysis of CO₂ emissions is considered an objective considering the environmental impacts of CO₂ emissions emitted from the process and equipment. From an economic perspective, maximization of profit is considered as one of the objectives. Profit can be calculated by the following equations (Eq 6-9).

$$\text{Net revenue} = \text{product sales} - \text{raw material cost} - \text{utility cost} - \text{energy cost} \quad (6)$$

$$\text{Utility cost} \left(\frac{\text{US\$}}{\text{h}} \right) = \text{cooling water cost} \left(\frac{\text{US\$}}{\text{h}} \right) + \text{steam cost} \left(\frac{\text{US\$}}{\text{h}} \right) \quad (7)$$

$$\text{Steam cost} \left(\frac{\text{US\$}}{\text{lb}} \right) = \frac{a_f (H_s - h_w)}{1000 \eta_B} \quad (8)$$

$$\text{Energy cost} \left(\frac{\text{US\$}}{\text{h}} \right) = \text{fuel gas cost} \left(\frac{\text{US\$}}{\text{h}} \right) + \text{electricity cost} \left(\frac{\text{US\$}}{\text{h}} \right) \quad (9)$$

Where a_f = fuel cost, (US \$/MMBtu),
 h_w = enthalpy of steam, (Btu/lb),
 h_w = enthalpy of boiler feedwater, (Btu/lb),
 η_B = overall boiler efficiency, fractional.

After the identification of objective functions, decision variables, and constraints, three cases are developed. Each case is executed independently, and the obtained results are discussed and analyzed in the results

Table 1
Decision variables & Constraints of the optimization problem.

Decision Variables & Constraints		Range
Decision Variables	Syngas molar flow (kmol/h)	35,000–50,000
	Reactor feed pressure (bar)	80–120
	Reactor 1 temperature (°C)	200–280
	Reactor 2 temperature (°C)	160–210
	By-product mass flow in column-3 (tonne/h)	2–5
Constraints	Ethanol mole fraction in methanol stream	Less than 0.01
	Methanol mole flow in the water to the treatment stream (kmol/h)	Less than 15

and discussion section. The following cases are considered in this optimization study:

Case 1. Maximize profit (US\$/y), and minimize total annual CO₂ emissions (kg/y)

Case 2. Maximize methanol production rate (kmol/h), and minimize total annual CO₂ emissions (kg/y)

Case 3. Maximize methanol production rate (kmol/h), minimize total annual CO₂ emissions (kg/y), and minimize total energy (kWh)

Such objectives which are conflicting in nature may show the impact on the methanol production process. The Case 1 and Case 2 studies deal with bi-objective optimization. In Case 3, tri objective optimization study is carried out by considering three sets of conflicting objectives. These objectives included maximization of methanol production (kmol/h), minimization of total annual CO₂ emissions (kg/y), and minimization of total energy (kWh). In Case 1, the profit and the annual CO₂ emissions are targeted considering the economics and environmental objectives. In Case 2, the process (methanol production rate) and environment (total annual CO₂ emissions) based objectives are targeted. Case 2 is further extended by adding a third conflicting objective of energy along with Case 2 objectives. Since profit is not explicitly considered in Case –2 and –3, it is analyzed as a calculated objective function, and compared using selected points in Case 3 results analysis. It must be noted that CO₂ emissions are calculated from both the process and the energy consumed by the equipment. Since energy is inherently considered in CO₂ emissions in the form of the equipment consumed energy, it is not considered in Case 1 and 2 study. However, the total energy-based objective is added to see its added impact as an objective in Case 3 in the tri-objective optimization study.

Population-based evolutionary algorithms are known for their ability to result in converged and diversified set of solutions also known as Pareto solutions. Several population-based algorithms are widely reported in the literature which includes, NSGA-II (Al-Aghbari et al., 2022), Multi-objective differential evolution (MODE) and its extensions (Gujarathi, 2023; Al Ani et al., 2021), Multi-objective Particle Swarm Optimization (MOPSO) (Hong et al., 2023), Multi-objective ant colony optimization algorithm (MOACO) (Liu and Liu, 2019), to name a few. NSGA-II is one of the most popular population-based evolutionary algorithms which holds three major advantages of having a fast non-dominated sorting approach, a fast crowded distance estimation procedure, and a simple crowded comparison operator (Srinivas and Deb, 1994). Differential evolution works on real numbers with crossover and recombination operators, and is further extended for solving MOO problems in the form of MODE and its extensions (Gujarathi, 2023). Particle swarm optimization (PSO) was inspired by the interaction of birds and insects and was modified to solve MOO problems (Hong et al., 2023). MOACO is a bio-inspired optimization algorithm (for solving MOO problems) based on simulating the behavior of natural ants that succeed in finding the shortest paths from their nest to food sources by communicating via a collective memory that consists of pheromone trails. These algorithms and many other evolutionary multi-objective optimization algorithms have shown potential in solving many complex and nonlinear MOO real-world problems.

NSGA-II is used in this study considering its wide range of applicability and success in -chemical and -other process industries to solve different problems (Palys et al., 2017; Chakraborti, 2004). The formulated cases were solved using NSGA-II and simulator interfacing approach. The objective function calculations were done in an Excel-based sheet using the NSGA-II algorithm and an interface code was built in VBA. The optimization runs were carried out for a population size of 100 and for a maximum of 100 generations. The flow chart for the interface between VBA, NSGA-II, and the simulator is shown in Fig. 2.

The detailed analysis of these conflicting objective functions and the impact of decision variables through three case studies consisting of two

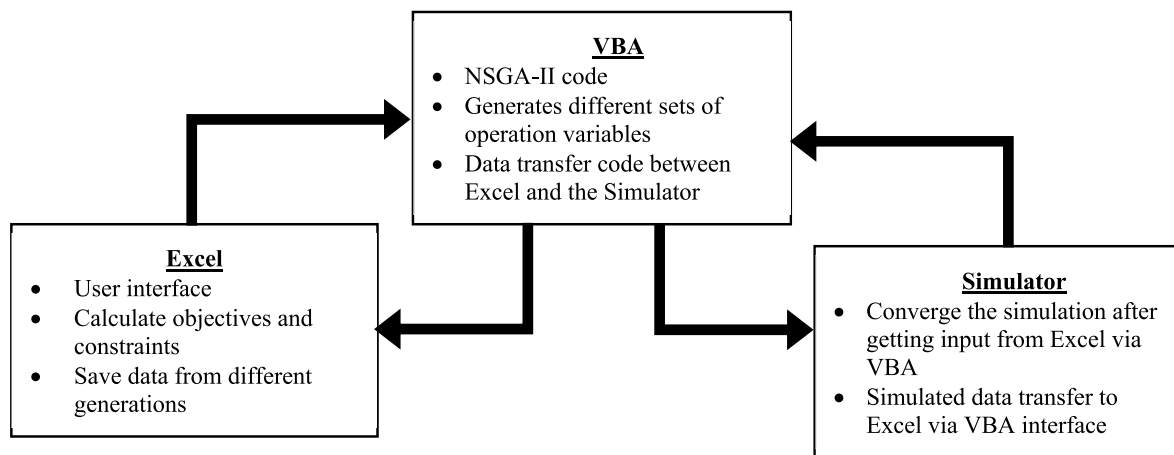


Fig. 2. Flow chart for the interface between VBA, NSGA-II, and the simulator.

bi-objective and one tri-objective optimization case is given in the next results and discussion section.

3. Results and discussion

3.1. Two-objective optimization study

3.1.1. Case 1: Maximizing profit and minimize total annual CO₂ emissions

In this case, the environmental and economic objective functions were considered for optimization study, where profit has been maximized while simultaneously minimizing total annual CO₂ emissions. Fig. 3 (a) shows the Pareto-optimal solution for the two-objective optimization study. Five different points are selected to compare the amount of CO₂ emissions for a particular value of profit. The objective function values along with the decision variables values of the selected points and the base case are presented in Table 2. Points A and E are extreme, point C is selected at the centre and points B and D are selected arbitrarily. The results show that CO₂ emissions increases with increasing profit. The decision variables values against the objective functions are shown in Fig. 3(b–i).

Fig. 3 (a) shows that Pareto solutions are discontinuous and do not seem to be a typical smooth Pareto front which may be possible for a complex real-world study with conflicting objectives. It must be noted that the solutions seem to be linearly varying specially between point B and point C, owing to high reactor 2 temperature (varying between 202 and 210 °C) and high value of variable by-product mass flow which vary in the range of 3.75–5 tons/h. Point A in Fig. 3 (a) shows the optimum operating conditions for a minimum amount of CO₂ emissions value which is 1.2354×10^9 kg/y when the profit is 7.538×10^8 US\$/y and the total calculated energy is 191190 kWh. Point E shows the operating conditions corresponding to the maximum profit of 10.364×10^8 US\$/y, with the corresponding total annual CO₂ emissions of 1.8313×10^9 kg/y and total calculated energy of 285750 kWh.

Table 2 shows that the base case has the lowest value of profit, and the highest value of CO₂ emissions compared to all the Pareto-optimal solutions obtained from the optimization. In the base case, the profit is equal to 3.44×10^8 US\$/y with the corresponding value of CO₂ emissions as 3.60×10^9 kg/y. In addition, the total energy for all selected points (from point A to point E) is lower than the base case value, which is 450392 kWh by 57.51%, 52.86%, 48.17%, 41.98%, and 36.49% respectively. Point E has resulted in the highest value of CO₂ emissions (1.8313×10^9 kg/y) among the selected points including the best optimal point. This is due to the decision variable syngas molar flow that attained the higher value (49735.08 kmol/h). Due to the same reason, point E also resulted in a higher total energy value (285750 kWh) and the highest profit (10.3645×10^8 US\$/y). Point A resulted in the

lowest values of both the set and calculated objectives. This could be due low value of syngas molar flow and the low value of reactor feed pressure.

Fig. 3(b–c) show how CO₂ emissions and profit change with syngas molar flow. Where a higher molar flow of syngas resulted in higher CO₂ emissions and profit. Increasing molar flow from the lower limit to the upper limit increases the profit approximately by 27.27%. According to Fig. 3(d–e) and reactor feed pressure values in Table 2, the optimal solutions of pressure were found between 80 and 112 bar, which covers the range of reactor feed pressure. Fig. 3 (f) shows that the reactor 1 temperature approached the higher bound of 233.54 °C for most of the solutions. Only two of the points (point D and point BOS) approached the lower value (218.71 °C and 220.98 °C) respectively. Fig. 3 (g) shows that the reactor 2 temperature values are scattered between 191.71 and 209.66 °C, approaching the upper bound of the decision variable. Fig. 3 (h–i) shows that the solutions for by-product mass flow in the column (Column-3) are scattered between its upper and lower bounds.

The motivation behind using Pareto ranking methods is to help the decision maker to choose the best optimal solution from the available set of non-dominated or Pareto solutions. For this purpose, the PROBID Pareto ranking method is used in this study. The PROBID method utilizes the following steps.

Step 1. Construction of normalized objective matrix by applying vector normalization.

Step 2. Construction of weighted normalized objective matrix by multiplying each column with its weight

Step 3. Determination of most, 2nd, 3rd, and ..., mth positive ideal solution (PIS) i.e., PIS_{S1}, PIS_{S2}, PIS_{S3}, ... and, PIS_{Sm} (where the mth PIS is also a most negative ideal solution (NIS)).

Step 4. Calculation of the Euclidean distance of each solution to each of the m ideal solutions and the mean solution.

Step 5. Determination of the overall positive-ideal distance

Step 6. Calculation of the ratio R_i (positive ideal/negative ideal) and then the performance score (P_i) of each solution

The solutions are ranked as per the performance score of individual solutions. Further detailed procedure of PROBID methods including formulas is available in the literature (Wang et al., 2021). Step 2 consists of assigning weights to individual objectives. For this purpose, the decision maker can add his or her assigned weights or can choose weights using the suggested weighing methods such as entropy weight, CRITIC, TOPSIS, SAW, GRA, MABAC, etc. Further details of weighing methods can be found in the literature (Wang et al., 2020). From the available weighing methods, the entropy weight method is utilized in this study

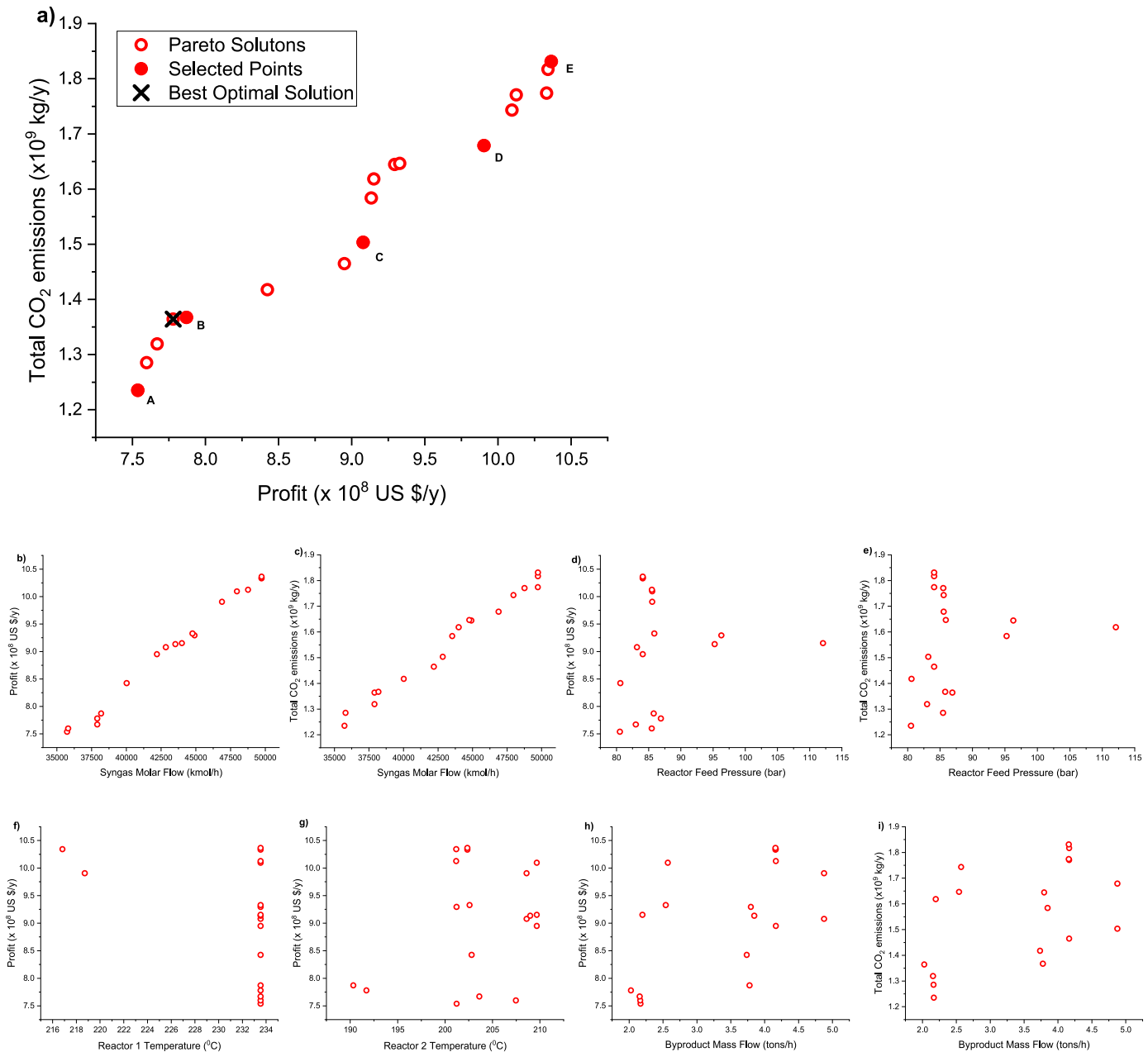


Fig. 3. Case 1 (a) optimized solution of maximizing profit (US\$/y) and minimizing CO₂ emissions (kg/y), (b–i) objectives function vs. decision variables.

Table 2

The objective functions results and the decision variables values for the base case and selected points from Fig. 3 (a).

Points	Decision Variables					Objective functions		
	Syngas Molar Flow (kmol/h)	Reactor Feed Pressure (bar)	Reactor 1 Temperature (°C)	Reactor 2 Temperature (°C)	By-product Mass Flow (tonne/h)	Profit x 10 ⁸ (US\$/y)	Total Annual CO ₂ Emissions x 10 ⁹ (kg/y)	Total Energy x 10 ⁵ (kWh) ^a
Base case	45544.58	92	250	190	4	3.44	3.60	4.50392
A	35733.27	80.52	233.54	201.22	2.168	7.5381	1.2354	1.9119
B	38186.93	85.78	233.54	190.33	3.778	7.8704	1.3674	2.1209
C	42843.30	83.18	233.54	208.60	4.876	9.0782	1.5035	2.3323
D	46885.73	85.56	218.71	208.59	4.876	9.9048	1.6788	2.6105
E	49735.08	84.09	233.54	202.35	4.160	10.3645	1.8313	2.8575
BOS	37903.86	86.88	220.98	191.71	2.027	7.7792	1.3644	2.1191

^a Calculated Value of objective function.

for all optimization cases. Under the entropy weight method, a weightage of 0.065 was assigned to profit objective, and a weightage of 0.9348 was assigned to the CO₂ emissions objective. The Pareto ranking results show that the best optimal solution (BOS) resulted in CO₂ emissions of 1.3644×10^9 with the corresponding value of profit as 7.779×10^8 US\$/y, with a calculated total energy of 211910 kWh.

3.1.2. Case 2: Maximizing methanol production rate and minimizing total annual CO₂ emissions

In this case, two conflicting objectives related to environment and process aspects i.e., minimization of the total annual CO₂ emissions while simultaneously maximizing methanol production rate are considered. The Pareto-optimal solutions are shown in Fig. 4 (a), which represents the minimum achievable amount of annual CO₂ emissions (kg/y) for a particular value of the rate of produced methanol (kmol/h). Fig. 4(b–i) show decision variables versus the studied objectives. Referring to Fig. 4 (a) the trend of minimization of the total annual CO₂ emissions versus maximization of methanol production rate can be divided into three intervals. The first interval is from A' to C' where CO₂ emissions increased slightly while methanol production rate increased

significantly. The second interval is between points C' and D' where no solutions were found. The third interval is between points D' and E' where CO₂ emissions increased sharply with a slight change in methanol production rate. The base case, Pareto, and the best optimal solution values of the objectives and operating variables for the selected points A', B', C', D' and E' are shown in Table 3.

The trade-off exists between these two objectives. Herewith in the first interval increase in methanol production rate from 9890.64 to 13900.80 kmol/h, CO₂ emissions increase from 1.021×10^9 to 1.70×10^9 kg/y. In the second interval, with an increase in methanol production rate from 13900.80 to 18494.58 kmol/h the corresponding increase in CO₂ emissions from 1.70×10^9 to 2.16×10^9 kg/y. This is consistent because the production of large amounts of methanol requires more amount of syngas. Similarly, with an increased amount of syngas, the amount of CO₂ consumption also increases since it is one of the main reactants. Owing to this, more electricity (for pumping and compression of the increased amount of fluid) and thus more burning of fuel causing more amount CO₂ emissions.

According to Table 3, the methanol production rate value of the base case is 12598.63 kmol/h located between point A' (9890.64 kmol/h)

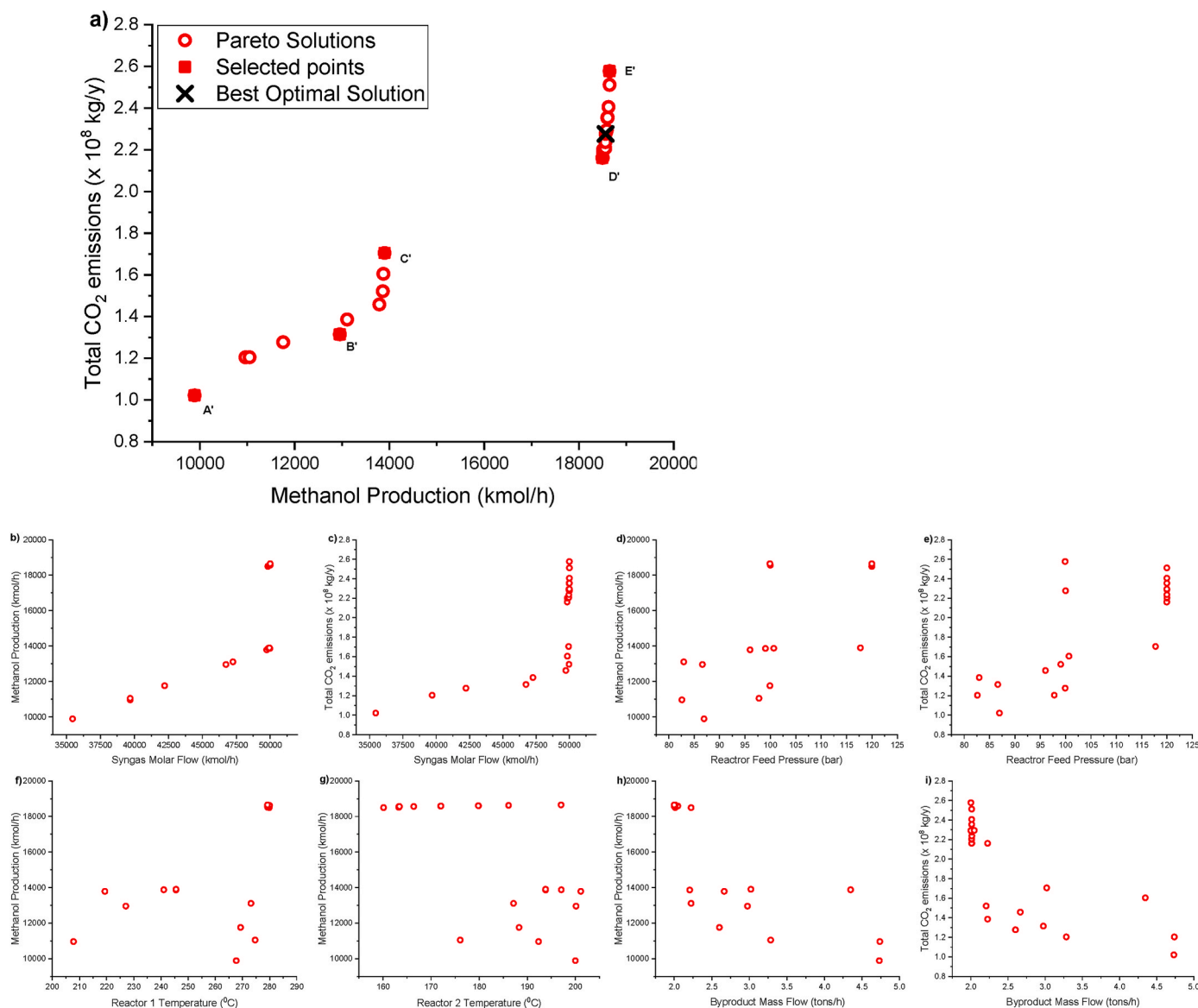


Fig. 4. Case 2 (a) optimized solution of maximizing methanol production rate (kmol/h) and minimizing CO₂ emissions (kg/y), (b–i) objectives function vs. decision variables.

Table 3

The objective functions results and the decision variables values for the base case and selected points from Fig. 4(a).

Points	Decision Variables					Objective Functions		
	Syngas Molar Flow (kmol/h)	Reactor Feed Pressure (bar)	Reactor 1 Temperature (°C)	Reactor 2 Temperature (°C)	By-product Mass Flow (tonne/h)	Methanol Production rate (kmol/h)	Total Annual CO ₂ Emissions x 10 ⁹ (kg/y)	Total Energy x 10 ⁵ (kWh) ^a
Base Case	45544.58	92	250	190	4	12598.63	3.60	4.50392
A'	35458.85	86.95	267.72	199.99	4.732	9890.637	1.02156	1.53254
B'	46740.79	86.64	227.07	200.13	2.975	12955.797	1.31471	1.95837
C'	49946.34	117.75	245.51	193.80	3.020	13900.804	1.70492	2.62616
D'	49828.97	119.99	279.88	160.13	2.226	18494.586	2.16129	3.41311
E'	49999.89	99.90	279.22	197.01	2.000	18647.811	2.57693	4.18224
BOS	49963.37	119.99	279.24	172.01	2	18576.18	2.29221	3.65475

^a Calculated value of the objective function.

and point B' (12955.797 kmol/h), however, with higher CO₂ emissions and total energy 3.60×10^9 kg/y and 450392 kWh. It is observed that the base case has the highest value of CO₂ emissions and total energy consumption compared to all the optimal solutions obtained from the optimization. Point A' resulted in the lowest value of both the set objective functions (methanol production rate and total annual CO₂ emissions) and also the calculated total energy-based objective. This could be due to the relatively lower values of syngas molar flow, reactor feed pressure, and relatively higher value of byproduct mass flow compared to the base case respective values. CO₂ emissions of points A' and E' are equal to 1.02×10^9 kg/y and 2.57×10^9 kg/y respectively where both are lower than the base case value which is 3.6×10^9 kg/y. It is worth noting that except for point A', all other points resulted in a higher amount of methanol production rate without sacrificing the values of total annual CO₂ emissions and total energy objectives compared to the base case. This is due to the higher syngas flow rate value (46740.79 kmol/h), higher temperature of a combination of reactors –1 and –2, and relatively lesser byproduct mass flow rate as shown in Table 3. In addition, the calculated total energy for all selected points (A' to E') is lower than the base case value (which is 450392.16 kWh) by 66.01%, 56.57%, 41.76%, 24.32%, and 7.26% respectively.

Syngas molar flow is one of the main variables which directly affects the methanol production rate. Looking for the interval A' to C' in Fig. 4 (a) and Table 3 the significant increase in the methanol production rate from 9890.64 to 13900.80 kmol/h is due to the increases in the syngas molar flow from 35458.85 kmol/h to approximately its upper limits. However, in the interval D' to E', it is clear that the syngas molar flow is at the maximum limit, and due to that, the production rate of methanol increases slightly by 0.82%. As shown in Fig. 4(d–e) the first interval bounded between points A' and C', the reactor feed pressure is scattered between 86 and 100 bar. For the second interval, all points converged in the upper bounds of 120 bar leading to a significant increase in the amount of CO₂ emissions.

According to Fig. 4 (f) and Table 3 the reactor 1 temperature is scattered between 207 °C and 275 °C, which is approximately from the upper to the lower bounds for the first interval. For the interval between point D' and point E', reactor 1 temperature is observed to be constant at the upper bound of 280 °C. Fig. 4 (g) shows that the reactor 2 temperature values are scattered between 175 °C and 201 °C for the first interval. For the interval between points D' and E' increasing reactor 2 temperature from its lower limit to 197 °C further increases the CO₂ emissions by 15.95%. For the first interval, the mass flow of the by-product stream is scattered between the respective bounds, and for the last interval maximum points are converged at the lower limits of 2 tonne/h as shown in Fig. 4(h–i). From the results of this case, it can be seen that, expecting higher production rates and operating the process at the upper bounds of the syngas molar flow rate, feed reactor pressure, and reactor 1 temperature cause a significant increase in CO₂ emissions. In the PROBD method, the weightage of 0.055775 was assigned to the methanol production rate objective and 0.944225 weightage was

assigned to the CO₂ emissions objective using the entropy weight method. The Pareto ranking results show that the best optimal solution (BOS) resulted in compromised values of all the objectives.

3.2. Three-objective optimization

3.2.1. Case 3: Maximizing methanol production rate and minimizing total annual CO₂ emissions and minimizing total energy (kWh)

Though in Case 2, two objectives namely, maximization of methanol production rate and minimization of total annual CO₂ emissions are considered, it is important to consider the same objectives along with equally important and conflicting objectives such as minimization of total energy. As the number of objectives in the given study increases the complexity of the problem increases whereas the results become more realistic towards the real-world operation of the process plant.

In this case, three objective functions were optimized simultaneously. Such type of optimization of the three objective functions is difficult compared with the two objectives optimization (Al-Mayyahi et al., 2011). The objectives of this case are minimizing total annual CO₂ emissions (kg/y), maximizing methanol production rate (kmol/h), and minimizing total energy (kWh). Fig. 5 (a) shows a 3-D plot of the optimal solutions for the three objective functions. Analyzing this case becomes more interesting by comparing it with the two objective cases that involved the same objective functions. Fig. 5 (b) compares the results of minimizing CO₂ emissions and maximizing methanol production rate from Case 2 and Case 3.

It can be noted from Fig. 5 (b) that the three objectives case is bordered by the two objectives case and it converges in a wider range. However, it is spread compared to Case 2 and this is due to the presence of the third objective, which is minimizing total energy. Table 4 compares Case 2 and Case 3 optimization results for some selected points in Fig. 5 (b).

Referring to Table 4 it can be observed that total energy values in Case 3 have been improved compared to Case 2 values. Where total energy consumption in P' reduced by 25.08% compared to solution P. This is due to including total energy as an objective function to be optimized. However, total energy values in Case 2 were not optimized and its reported value is the calculated value based on the 2-objective optimization of the methanol production rate and total annual CO₂ emissions case study. Similarly, the total energy values for points Q' to T' are improved by 22.69%, 23.87%, 23.41%, and 24.07% respectively. Similarly, while moving from point P' to R' the annual CO₂ emissions values increase by 1.01%, 0.93%, and 0.042% respectively. For points S' and T' the annual CO₂ emissions values decrease by 0.43% and 0.59%. It is worth noting that the methanol production rate is highest in the base case compared to all points of Case 3 results. This is due to the low value of syngas molar flow and reactor feed pressure. However, except for point P, while moving from point Q to point T (in Case 2), the methanol production rate increases by 2.83%, 10.13%, 47.33%, and 48.01%. The best optimal solution resulted in an improved value of methanol

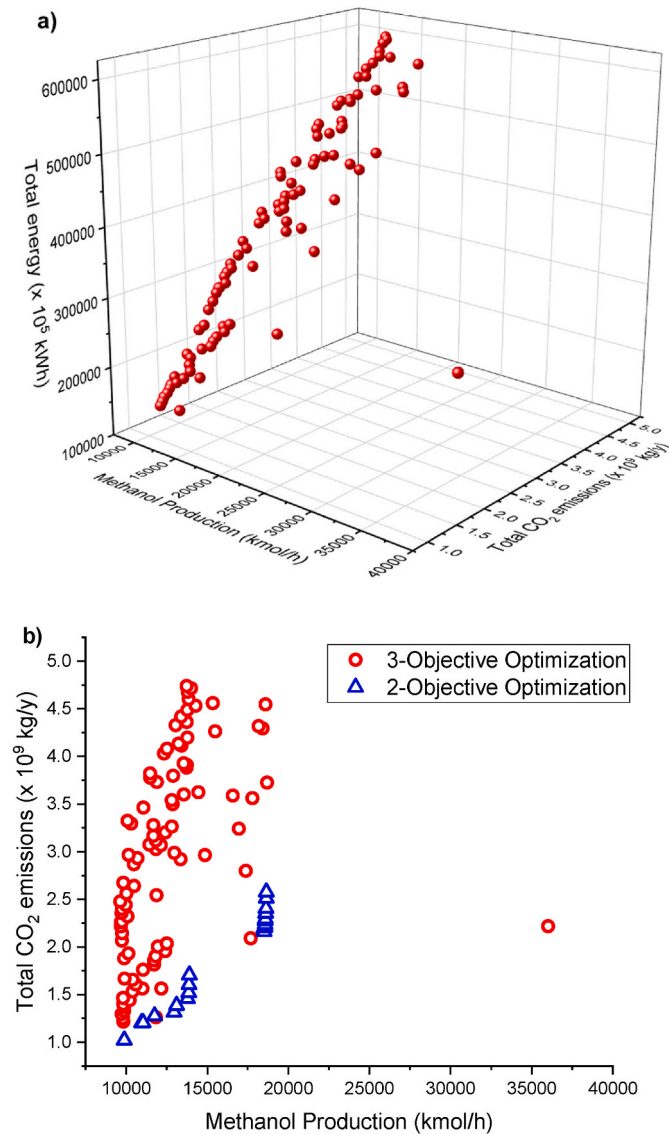


Fig. 5. Case 3 (a) optimized solution of maximizing methanol production rate (kmol/h), minimizing total energy (kWh) and minimizing CO₂ emissions (kg/y), (b) a comparison between Case 2 and Case 3.

production rate by 47.45%. Considering the entropy weight method, the weight values for the PROBIT method for Case 3 are 0.064401, 0.184903, and 0.750697 for the objectives of methanol production rate, total annual CO₂ emissions, and total energy respectively. Due to the involvement of the third conflicting objective i.e., total energy minimization, the maximum CO₂ emissions value in Case 3 is 4.7389×10^9 kg/y which is 83.89% higher than the highest value of CO₂ emissions obtained in Case 2 results. Similarly, a single point in Case 3 is deviated from all other points in terms of methanol production rate and approached a value of 36015.49 kmol/h with an increase of 93.13% compared to the maximum methanol production rate obtained in Case 2 results. The obtained total energy objective values in Case 2 (calculated) and Case 3 are 153254.18 kWh and 136232.56 kWh, showing significant improvement in Case 3 results. The key decision variables responsible for the higher value of CO₂ emissions in Case 3 are reactor feed pressure, reactor 1 temperature, reactor 2 temperature, by-product mass flow rate, and their corresponding values in Case 3 along with corresponding values in Case 2 (shown in bracket) are 82.01 (99.90) bar, 208.65 (279.22) °C, 206.34 (197.01) °C, 3.97 (2) tonne/h, respectively. Similarly, the key decision variables responsible for the highest value of methanol production rate in Case 3 compared to the highest methanol production rate in Case 2 are reactor feed pressure, reactor 1 temperature, reactor 2 temperature, by-product mass flow rate, with their corresponding values of 83.83 bar, 272.73 °C, 206.33 °C, 2.42 tonne/h in Case 3, and 99.89 bar, 279.22 °C, 197.01 °C, 2.01 tonne/h in Case 2 respectively. The average value of CO₂ emissions in Case 2 and Case 3 are 2.033×10^9 kg/y and 2.9161×10^9 kg/y, whereas the average value of methanol production rate in Case 2 and Case 3 are 17071.82 kmol/h and 12640.76 kmol/h. The average value of total energy objective in Case 2 (calculated) and Case 3 are 320563.76 kWh and 360233.13 kWh.

Fig. 6 shows the process and equipment based annual CO₂ emissions for selected points from Case 1, Case 2 and Case 3 solutions. The process based annual CO₂ emissions are the actual amount of CO₂ as emitted through the process streams. The equipment based annual CO₂ emissions are calculated from the CO₂ generated from the burnt fuel to produce the electricity consumed by the respective equipment. The equipment based annual CO₂ emissions amounts are much larger than the CO₂ emissions through the process.

4. Conclusions

Multi-objective optimization for the large scale methanol Lurgi two-steps process was conducted using a NSGA-II optimization approach. Conflicting objective functions related to process, environment, and economic aspects were used to formulate the optimization problem. In

Table 4

Comparison of the selected optimized solution for two objective (Case 2) and three objectives (Case 3) optimization case studies (From Fig. 5(b)).

Type	Point	Objective Functions				Decision Variables				
		Methanol Production rate (kmol/h)	Total Annual CO ₂ Emissions x 10 ⁹ (kg/y)	Total Energy x 10 ⁵ (kWh)	Profit x 10 ⁸ (US \$/y) ^a	Syngas Molar Flow (kmol/h)	Reactor Feed Pressure (bar)	Reactor 1 Temperature (°C)	Reactor 2 Temperature (°C)	By-product Mass Flow (tonne/h)
Base Case		12598.63	3.60	4.50392	3.44	45544.58	92	250	190	4
Two Objective Optimization	P	10963.78	1.20408	1.81879	2.618	39698.66	82.60	207.78	192.35	4.74
	Q	12955.80	1.31471	1.95837	3.081	46740.79	86.64	227.07	200.13	2.97
	R	13875.39	1.60509	2.44851	3.470	49852.08	100.66	241.02	197.07	4.35
	S	18561.84	2.27591	3.61638	4.002	49999.66	99.99	279.26	163.42	2.01
	T	18647.81	2.57693	4.18224	3.517	49999.89	99.90	279.22	197.01	2.00
Three Objective Optimization	BOS	18576.18	2.29221	3.65475	4.084	49963.37	119.99	279.24	172.01	2
	P'	9832.13	1.21631	1.36250	2.258	35428.44	87.96	262.73	205.55	2.60
	Q'	9880.15	1.32697	1.51400	2.214	35418.34	83.85	201.56	206.33	2.47
	R'	10597.88	1.60578	1.86396	2.590	38020.30	87.91	273.07	188.58	4.92
	S'	9742.29	2.26607	2.76956	2.686	35383.56	88.46	266.11	162.57	4.92
	T'	10024.73	2.56158	3.17541	2.459	36144.47	83.85	263.07	188.58	2.47
	BOS	11070.62	3.46268	4.36612	2.813	39895.30	88.46	203.65	186.28	2.13

^a Calculated value of the objective function.

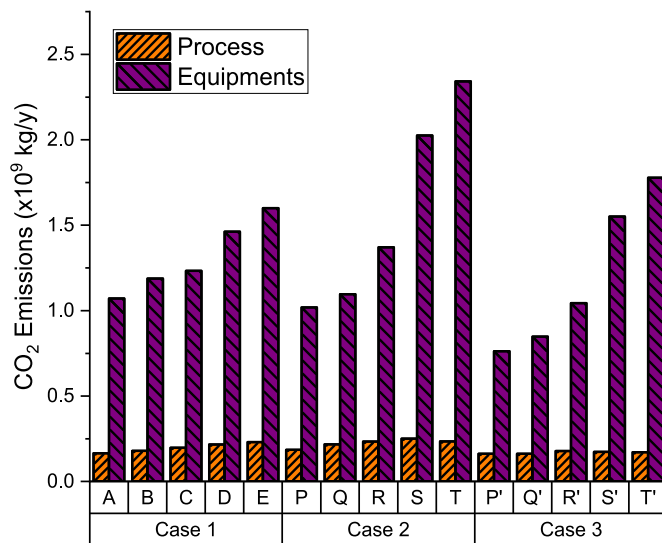


Fig. 6. Process and Equipment based annual CO₂ emissions for selected points from different cases.

addition, five different decision variables were chosen for the optimization study. The main findings from the analysis of results can be summarized as follows:

After comparing the result of the optimized cases with the base case (un-optimized case), it is evident that the proposed optimization approach has shown improvement over the base case results. **Case 1**, CO₂ emissions decreased by 65.83%, methanol production rate increased by 10.11%, and total energy reduced by 57.55%. Maximizing

the profit and minimizing CO₂ emissions (**Case 1**) could be achieved by increasing syngas molar flow, operating the reactor at 80–85 bar, and setting reactor 2 temperature in the range of 200–210 °C. Maximum methanol production rate (**Case 2**) could be achieved by increasing syngas molar flow, operating the reactor at a higher specified pressure, and setting by-product mass flow of column-3 at 2 tonne/h. Pareto solutions obtained in three objective cases (**Case 3**) are wider and more scattered than two-objective case solutions due to the presence of the third conflicting objective.

CRediT authorship contribution statement

Mallak Al Riyami: Writing – original draft, Software, Formal analysis. **Swaprabha P. Patel:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Ashish M. Gujarathi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rashid Al-Hajri:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Appendix A1

Four main reactions occurring in methanol reactors are given through equations (1)–(4). The design details for methanol reactors are given in **Table A1**:

Table A1
Design details for methanol reactors.

Parameter	Reactor	
	Reactor-1	Reactor-2
Tube diameter	11 cm	11 cm
Tube length	6.2 m	6.2 m
Number of tubes	4250	6500
Reactor type	Reactor constant thermal fluid temperature	Reactor with counter-current thermal fluid
Thermal fluid temperature	271 °C	255 °C
Bed voidage	0.38	0.38
Particle density	1960 kg/cum	1960 kg/cum

Frequency factor (k_i), activation energy (E_i), and reference temperature (T_0) for four reactions are given in **Table A2**.

Table A2
Frequency factor, activation energy, and reference temperature for four reactions

Reactions	Frequency factor (k_i)	Activation energy (E_i) (kcal/mol)	Reference temperature (T_0) (°C)
Reaction 1	7.07034	−8.76469	228.42
Reaction 2	0.00165	22.6342	228.42
Reaction 3	1^{-12}	19.467	228.42
Reaction 4	4.2×10^{-11}	18.66	228.42

The rates of reactions are as follows:

$$r_1 = k_1 (p_{CO_2} p_{H_2}) \quad (A.1)$$

$$r_2 = k_2 (p_{CO_2}) \quad (A.2)$$

$$r_3 = k_3 (p_{CO} p_{H_2}^{1.5}) \quad (A.3)$$

$$r_4 = k_4 (p_{MeOH}^{0.55}) \quad (A.4)$$

Column configurations:

Column-1:

Calculation type = equilibrium, Number of stages = 10, stage1/condenser pressure = 86.5 bar.

Column-2:

Calculation type = equilibrium, Number of stages = 27, stage1/condenser pressure = 1.35799 bar, column pressure drop = 0.344738 bar, Design specs: Distillate rate = 3 tonne/h, Reboiler duty = 45.36 Gcal/h.

Column-3:

Calculation type = equilibrium, Number of stages = 50, stage1/condenser pressure = 1.35799 bar, column pressure drop = 0.344738 bar, Design specs: Reflux ratio = 1.5, Reboiler duty = 262.08 Gcal/h.

The list of the equipment shown in Fig. 1 and its role is shown in Table A3.

Table A3
Equipment list and role

Sr. No.	Name	Role of device
1	Reactor-1	Methanol Reactor Stage –1
2	Reactor-2	Methanol Reactor Stage –2
3	Column-1	Scrubber
4	Column-2	Light ends column
5	Column-3	Methanol Product column
6	Tank-1	Crude methanol storage tank
7	Vessel-1	Knock out drum
8	Vessel-2	Crude methanol flash drum
9	Vessel-3	Flash column
10	Comp-1	Synthesis gas compressor
11	Comp-2	Recycle gas compressor
12	Comp-3	Synthesis gas compressor level 2
13	Comp-4	Column-1 product compressor
14	Pump-1	Column-2 feed pump
15	Pump-2	Column-3 feed pump
16	Pump-3	Recycle pump
17	Pump-4	Column-3 bottoms pump
18	HEX-1	Reactor inter-heater
19	HEX-2	Interchange heat exchanger
20	HEX-3	Syngas compressor cooler
21	HEX-4	Interchange heat exchanger
22	HEX-5	Partial condenser
23	HEX-6	Reactor product air cooler
24	HEX-7	Reactor product cooler
26	HEX-8	Column-3 Product Cooler (cross)
27	HEX-9	Column-2 feed heater (cross)

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